

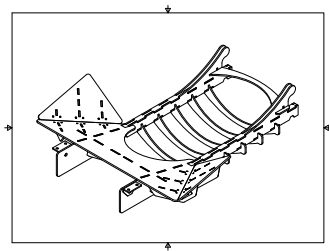
Maker Portfolio

FIRST Robot Chute

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Wilson Robotics Club

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Purpose

FIRST Robotics Competition

The goal for the 2022 FRC game was to create a robot that could gather large foam balls from the ground and launch them into a central hoop for varying amounts of points, competing against many other teams

The robot

The overall design consisted of three main assemblies: the **chassis**, **ball intake**, **ball shooter**, and the **receiving chute**

My part

My role on the team was to design and build the **receiving chute** assembly, which funneled balls from the intake to the shooter

Constraints

- Must be manufactured with available tools and materials
- Must be able to survive impacts from the ball
- Must fit and fasten to the robot's existing frame
- Must not obstruct other assemblies or functions
- Must reliably guide ball from intake to shooter

My Design

Using Autodesk Fusion 360

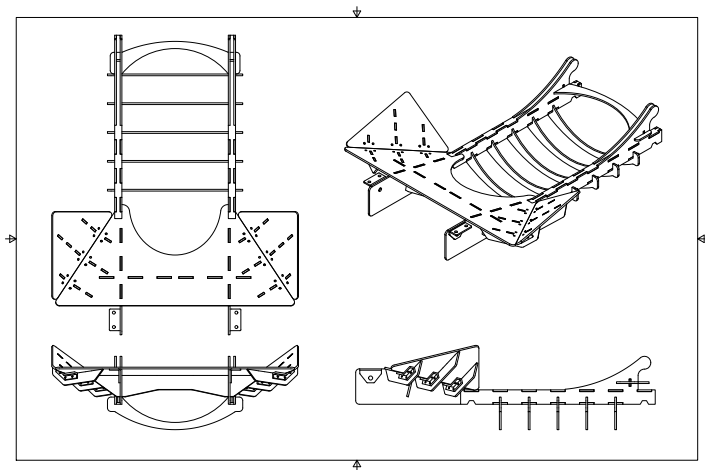


Figure: Orthographic projections of assembly

Machining

Materials

Parts cut out of 1/8" Masonite sheets; a cheap, relatively strong, and easily machinable material

Tools

Our Epilog Lasers Zing 24 laser cutter needed extensive cleaning and setup to become fully operational

Calibration

Extensive trial and error required to avoid burnt edges or incomplete cuts



Figure: Test squares arranged by laser frequency (horizontal) and speed (vertical)

Parts

Laid out before construction

Figure: Exploded view of 3D model

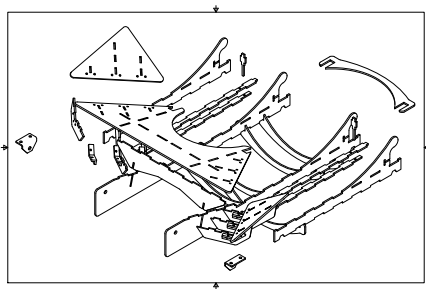


Figure: Laser cut and 3D printed parts

Assembly

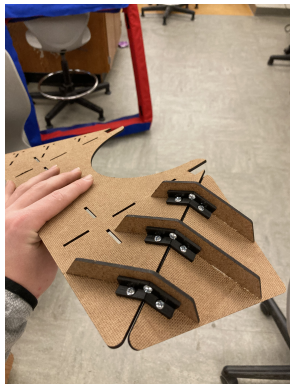


Figure: Ramp, sides, and braces glued together; fastened by 3D printed brackets

Box joints

Tight fitting box joints secured with wood glue for durable and lightweight connections

Bracing

Perpendicular braces designed to prevent deflection in main ramp and rails

3D printed brackets

Sides of ramp reinforced with 3D printed brackets bolted into ramp, its sides, and braces

Integration

Fastening

Chute attached to frame with zip-ties to avoid protruding bolt ends

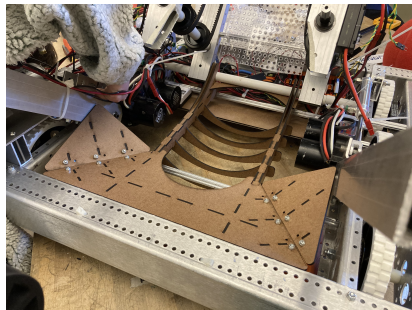
Fit

Assembly engages with frame's aluminum hex spars and aligns with edges of frame and intake

Function

Ball rolls down ramp and is guided between the two rails, which curve up into shooter

Figure: Full assembly after installation in robot



Failure

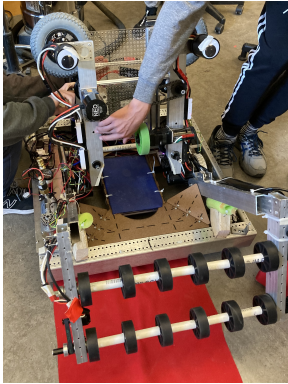


Figure: Robot at competition with various last-minute alterations

Broken ribs

During hard impacts the ball was able to knock ribs off the ramp; pry apart its rails

Obstruction

Horizontal spars prevented ball from rolling perfectly on rails

Solution

Design flaws remedied by placement of flexible sheet of felt between rails to provide elastic resistance against ball's impact

Robot triumph

In competitions our robot was successful at gathering balls and scoring points, winning various rounds

Team effort

Thanks to good collaboration and communication with my team, my assembly was able to play its part well with the rest of the system

Skill development

This process of design, construction, trial, and error provided extensive growth for my CAD and CNC machining skills, and was one of my first big collaborative engineering projects